

ELEMENTARY MENSURATION – Question Wise Complete Formula Sheet

Q1. Volume of Cuboid

Formula Used

Volume of Cuboid = Length $\times$ Breadth $\times$ Height

where:

- Length = longer side
- Breadth = shorter side
- Height = vertical side

Q2. Total Surface Area of Cuboid

Formula Used

Total Surface Area = $2[(\text{Length} \times \text{Breadth}) + (\text{Breadth} \times \text{Height}) + (\text{Height} \times \text{Length})]$

Q3. Diagonal of Cuboid

Formula Used

$(\text{Diagonal})^2 = (\text{Length})^2 + (\text{Breadth})^2 + (\text{Height})^2$

$$\text{Diagonal} = \sqrt{[(\text{Length})^2 + (\text{Breadth})^2 + (\text{Height})^2]}$$

Q4. Volume of Cube

Formula Used

$$\text{Volume of Cube} = (\text{Side})^3$$

Q5. Total Surface Area of Cube

Formula Used

$$\text{Total Surface Area} = 6 \times (\text{Side})^2$$

Q6. Diagonal of Cube

Formula Used

$$\text{Diagonal} = \text{Side} \times \sqrt{3}$$

Q7. Curved Surface Area of Cylinder

Formula Used

$$\text{Curved Surface Area} = 2\pi \times \text{Radius} \times \text{Height}$$

Q8. Total Surface Area of Cylinder

Formula Used

Total Surface Area = $2\pi \times \text{Radius} \times (\text{Radius} + \text{Height})$

Q9. Volume of Cylinder

Formula Used

Volume = $\pi \times (\text{Radius})^2 \times \text{Height}$

Q10. Curved Surface Area of Cone

Formula Used

Curved Surface Area = $\pi \times \text{Radius} \times \text{Slant Height}$

Q11. Total Surface Area of Cone

Formula Used

Total Surface Area = $\pi \times \text{Radius} \times (\text{Radius} + \text{Slant Height})$

Q12. Volume of Cone

Formula Used

$$\text{Volume} = (1/3) \times \pi \times (\text{Radius})^2 \times \text{Height}$$

Q13. Sphere Formula

Surface Area

$$\text{Surface Area} = 4\pi \times (\text{Radius})^2$$

Volume

$$\text{Volume} = (4/3) \times \pi \times (\text{Radius})^3$$

Q14. Slant Height of Cone

Radius

$$\text{Radius} = \text{Diameter} / 2$$

Pythagoras Theorem

$$(\text{Slant Height})^2 = (\text{Radius})^2 + (\text{Height})^2$$

$$\text{Slant Height} = \sqrt{[(\text{Radius})^2 + (\text{Height})^2]}$$

Q15. Percentage of Wood Wasted

Volume of Cone

$$\text{Volume} = (1/3) \times \pi \times (\text{Radius})^2 \times \text{Height}$$

Volume of Cuboid

$$\text{Volume} = \text{Length} \times \text{Breadth} \times \text{Height}$$

Percentage Wasted

$$\text{Percentage Wasted} = [(\text{Volume of Cuboid} - \text{Volume of Cone}) / \text{Volume of Cuboid}] \times 100$$

Q16. Total Surface Area of Cone

Slant Height

$$(\text{Slant Height})^2 = (\text{Radius})^2 + (\text{Height})^2$$

Total Surface Area

$$\text{Total Surface Area} = \pi \times \text{Radius} \times (\text{Radius} + \text{Slant Height})$$

Q17. Volume of Frustum

Circumference Formula

$$\text{Circumference} = 2\pi \times \text{Radius}$$

$$\text{Radius} = \text{Circumference} / (2\pi)$$

Volume of Frustum

$$\text{Volume} = (1/3) \times \pi \times \text{Height} \times [(\text{Radius1})^2 + (\text{Radius2})^2 + (\text{Radius1} \times \text{Radius2})]$$

Q18. Cube Surface Area to Volume

Surface Area of Cube

$$\text{Surface Area} = 6 \times (\text{Side})^2$$

Volume of Cube

$$\text{Volume} = (\text{Side})^3$$

Q19. Cubes Melted into New Cube

Volume of Cube

$$\text{Volume} = (\text{Side})^3$$

Total Volume

$$\text{New Cube Volume} = \text{Volume1} + \text{Volume2} + \text{Volume3}$$

Side of New Cube

$$\text{Side} = \sqrt[3]{(\text{Total Volume})}$$

Q20. Smaller Cubes from Big Cube

Volume of Cube

$$\text{Volume} = (\text{Side})^3$$

Number of Cubes

$$\text{Number of Smaller Cubes} = \text{Volume of Big Cube} / \text{Volume of Small Cube}$$

or

$$\text{Number} = (\text{Big Side} / \text{Small Side})^3$$

Q21. Weight of Iron Pipe

Outer Radius

$$\text{Outer Radius} = \text{Inner Radius} + \text{Thickness}$$

Volume of Pipe

$$\text{Volume} = \pi \times \text{Length} \times [(\text{Outer Radius})^2 - (\text{Inner Radius})^2]$$

Weight

$$\text{Weight} = \text{Volume} \times \text{Weight per cubic cm}$$

Q22. Sphere Drawn into Wire

Volume of Sphere

$$\text{Volume} = \frac{4}{3} \times \pi \times (\text{Radius})^3$$

Volume of Wire

$$\text{Volume} = \pi \times (\text{Radius})^2 \times \text{Length}$$

Volume Relation

$$\text{Volume of Sphere} = \text{Volume of Wire}$$

Q23. Cylinder Melted into Sphere

Volume of Cylinder

$$\text{Volume} = \pi \times (\text{Radius})^2 \times \text{Height}$$

Volume of Sphere

$$\text{Volume} = \frac{4}{3} \times \pi \times (\text{Radius})^3$$

Equality of Volumes

$$\text{Volume of Cylinder} = \text{Volume of Sphere}$$

Q24. Cylinder Melted into Spheres

Volume of Cylinder

$$\text{Volume} = \pi \times (\text{Radius})^2 \times \text{Height}$$

Volume of Sphere

$$\text{Volume} = (4/3) \times \pi \times (\text{Radius})^3$$

Number of Spheres

$$\text{Number} = \text{Volume of Cylinder} / \text{Volume of One Sphere}$$

Q25. Greatest Sphere inside Cone

Slant Height

$$(\text{Slant Height})^2 = (\text{Radius})^2 + (\text{Height})^2$$

Radius of Sphere

$$\text{Radius of Sphere} = (\text{Radius of Cone} \times \text{Height of Cone}) / (\text{Radius of Cone} + \text{Slant Height})$$

Q26. Ratio of Volumes of Spheres

Surface Area of Sphere

$$\text{Surface Area} = 4\pi \times (\text{Radius})^2$$

Volume of Sphere

$$\text{Volume} = \frac{4}{3} \times \pi \times (\text{Radius})^3$$

Ratio Rule

$$\text{If Surface Area Ratio} = a : b$$

$$\text{Then Radius Ratio} = \sqrt{a} : \sqrt{b}$$

$$\text{Then Volume Ratio} = (\sqrt{a})^3 : (\sqrt{b})^3$$

Q27. Curved Surface Area of Cylinder

Formula

$$\text{Curved Surface Area} = 2\pi \times \text{Radius} \times \text{Height}$$

New Radius

$$\text{New Radius} = 3 \times \text{Radius}$$

New Height

$$\text{New Height} = \frac{1}{3} \times \text{Height}$$

Q28. Cone Formed by Rotating Triangle

Volume of Cone

$$\text{Volume} = \frac{1}{3} \times \pi \times (\text{Radius})^2 \times \text{Height}$$

Slant Height

$$(\text{Slant Height})^2 = (\text{Radius})^2 + (\text{Height})^2$$

Curved Surface Area

$$\text{Curved Surface Area} = \pi \times \text{Radius} \times \text{Slant Height}$$

Total Surface Area

$$\text{Total Surface Area} = \pi \times \text{Radius} \times (\text{Radius} + \text{Slant Height})$$

Q29. Percentage Increase in Cuboid Volume

Original Volume

$$\text{Volume} = \text{Length} \times \text{Breadth} \times \text{Height}$$

New Volume

$$\text{New Volume} = (2 \times \text{Length}) \times (3 \times \text{Breadth}) \times (4 \times \text{Height})$$

Percentage Increase

$$\text{Percentage Increase} = [(\text{New Volume} - \text{Original Volume}) / \text{Original Volume}] \times 100$$

Q30. Length Made Four Times

Original Volume

$$\text{Volume} = \text{Length} \times \text{Breadth} \times \text{Height}$$

New Volume

$$\text{New Volume} = (4 \times \text{Length}) \times \text{Breadth} \times \text{Height}$$

Percentage Increase

$$\text{Percentage Increase} = [(\text{New Volume} - \text{Original Volume}) / \text{Original Volume}] \times 100$$

Q31. Edge of Cube Increased by 10%

Volume of Cube

$$\text{Volume} = (\text{Side})^3$$

New Side

$$\text{New Side} = 1.10 \times \text{Side}$$

New Volume

$$\text{New Volume} = (1.10)^3 \times \text{Original Volume}$$

Percentage Increase

$$\text{Percentage Increase} = [(\text{New Volume} - \text{Original Volume}) / \text{Original Volume}] \times 100$$